

Mathematical Methods for Finance

Prerequisite Crash Course for Lectures 1–2

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Role of this file

The complete prerequisite file is the reference. These slides are the live classroom version: short theory, many examples, fast exercises.

- ▶ Use during the last 50 minutes of Lecture 1 and the full Lecture 2.
- ▶ The diagnostic test will be corrected later and used to tune future recalls.
- ▶ The objective is to reactivate tools, not to repeat the whole undergraduate programme.

Structure of the first two lectures

Time	Activity
Lecture 1 - block 1	Website, materials, programme, assessment rules.
Lecture 1 - block 2	Initial diagnostic test.
Lecture 1 - block 3	Prerequisites, block A.
Lecture 2 - block 1	Prerequisites, block B.
Lecture 2 - block 2	Prerequisites, block C.
Lecture 2 - block 3	Prerequisites, block D.

Four prerequisite blocks

Block A	Sets, real line, absolute value, distances, functions.
Block B	Limits, continuity, derivatives, optimization, integrals.
Block C	Vectors, matrices, linear systems, functions of several variables.
Block D	Descriptive statistics, probability, random variables, basic inference.

How to work on each exercise

1. Identify the object: set, function, vector, matrix, event, random variable.
2. Write the relevant formula or definition.
3. Check the domain, dimensions or assumptions.
4. Compute only what is necessary.
5. End with a short interpretation.

Common mistake

Starting computations before checking the domain, dimensions or assumptions.

Sets, functions and distances

Objects we will meet repeatedly

Object	Notation	Question
Set	A, B, Ω	Which elements are allowed?
Function	$f : X \rightarrow Y$	What output corresponds to each input?
Sequence	(x_n)	Does it approach a limit?
Vector	$x \in \mathbb{R}^n$	Which coordinates does it have?
Matrix	$A \in \mathbb{R}^{m \times n}$	Which operation does it represent?
Random variable	X	Which numerical outcome is uncertain?

Set operations

For sets A and B : $A \cup B$ means in A or in B ; $A \cap B$ means in both; $A \setminus B$ means in A but not in B .

Student task

Let $A = \{1, 2, 4, 6\}$ and $B = \{2, 3, 6\}$. Compute $A \cup B$, $A \cap B$, $A \setminus B$.

Solution

$$A \cup B = \{1, 2, 3, 4, 6\}, \quad A \cap B = \{2, 6\}, \quad A \setminus B = \{1, 4\}.$$

The difference $A \setminus B$ keeps only the elements of A that are not in B .

Bounds

A set $E \subseteq \mathbb{R}$ is bounded above if there is M such that $x \leq M$ for all $x \in E$. The least upper bound is $\sup E$. The greatest lower bound is $\inf E$.

Student task

For $E = (-2, 5]$, find $\inf E$, $\sup E$, minimum and maximum.

Solution

$$\inf E = -2, \quad \sup E = 5.$$

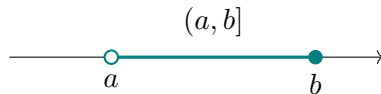
There is no minimum because $-2 \notin E$. There is a maximum and $\max E = 5$ because $5 \in E$.

Open

A set is open if around each of its points we can draw a small interval still contained in the set.

Closed

A set is closed if it contains its boundary points.



Student task

In $\mathbb{R}$, classify the following sets as open, closed, both, or neither:

$$(0, 1), \quad [0, 1], \quad (0, 1], \quad \mathbb{R}.$$

Quick classification: solution

Solution

$(0, 1)$: open, $[0, 1]$: closed, $(0, 1]$: neither, $\mathbb{R}$: both.

The real line has no boundary point outside itself, so it is both open and closed.

Absolute value: the key prerequisite for distance

Absolute value

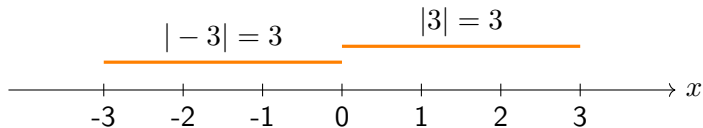
$$|x| = \begin{cases} x, & x \geq 0, \\ -x, & x < 0. \end{cases}$$

It measures the distance of x from zero.

Key idea

The expression $|x - y|$ measures the distance between x and y on the real line.

Absolute value as a picture



Memory rule

Absolute value removes the sign but keeps the magnitude.

Useful absolute value rules

Rule	Meaning
$ x \geq 0$	distance cannot be negative
$ x = 0 \Leftrightarrow x = 0$	only zero has zero distance from zero
$ xy = x y $	magnitudes multiply
$ x + y \leq x + y $	triangle inequality
$ x - y < r$	x is within distance r from y

Absolute value inequalities

Key idea

The inequality $|x - a| < r$ means x is between $a - r$ and $a + r$.

$$|x - a| < r \iff a - r < x < a + r.$$

Student task

Solve $|x - 2| < 3$ and $|x + 1| \leq 4$.

Solution

$$|x - 2| < 3 \iff -3 < x - 2 < 3 \iff -1 < x < 5.$$

$$|x + 1| \leq 4 \iff -4 \leq x + 1 \leq 4 \iff -5 \leq x \leq 3.$$

Distance on the real line

Usual distance

On $\mathbb{R}$, the usual distance is $d(x, y) = |x - y|$.



Open ball in $\mathbb{R}$

With $d(x, y) = |x - y|$, the open ball centered at a with radius r is

$$B(a, r) = \{x : |x - a| < r\} = (a - r, a + r).$$

Student task

Find $B(2, 0.5)$ and $B(-1, 3)$.

Solution

$$B(2, 0.5) = (1.5, 2.5), \quad B(-1, 3) = (-4, 2).$$

An open ball on the real line is simply an open interval.

Functions: first check the domain

Function

A function $f : X \rightarrow Y$ assigns to each $x \in X$ exactly one value $f(x) \in Y$.

Common mistake

Before solving, differentiating or composing a function, check where the formula is defined.

Student task

Find the natural domain of the following functions:

$$f(x) = \sqrt{x-1}, \quad g(x) = \frac{1}{x^2-4}, \quad h(x) = \log(3-x).$$

Solution

$$f : x - 1 \geq 0 \Rightarrow x \geq 1.$$

$$g : x^2 - 4 \neq 0 \Rightarrow x \neq \pm 2.$$

$$h : 3 - x > 0 \Rightarrow x < 3.$$

The domains are $[1, +\infty)$, $\mathbb{R} \setminus \{-2, 2\}$, and $(-\infty, 3)$.

Composition

If g sends x into a value accepted by f , then $(f \circ g)(x) = f(g(x))$.

Student task

Let $f(x) = x^2 - 1$ and $g(x) = 2x + 3$. Compute $(f \circ g)(x)$ and $(g \circ f)(x)$.

Solution

$$(f \circ g)(x) = f(2x + 3) = (2x + 3)^2 - 1 = 4x^2 + 12x + 8.$$

$$(g \circ f)(x) = g(x^2 - 1) = 2(x^2 - 1) + 3 = 2x^2 + 1.$$

Usually $f \circ g \neq g \circ f$.

Injective function

A function is injective if different inputs give different outputs. Only injective functions can be inverted on their domain.

Student task

Is $f(x) = x^2$ injective on $\mathbb{R}$? Is it injective on $[0, +\infty)$?

Solution

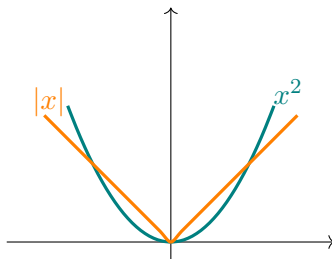
On $\mathbb{R}$, f is not injective because $f(1) = f(-1) = 1$. On $[0, +\infty)$, it is injective because x^2 is increasing there. Its inverse on $[0, +\infty)$ is $f^{-1}(y) = \sqrt{y}$.

Convexity: the shape matters

Convex function

A function is convex if the segment joining two points of its graph lies above the graph.

- ▶ x^2 is convex.
- ▶ $|x|$ is convex but not differentiable at zero.



Five-minute check

1. Solve $|x - 4| \leq 2$.
2. Find the domain of $f(x) = \sqrt{2 - x}$.
3. Compute $(f \circ g)(x)$ for $f(x) = |x|$, $g(x) = x - 3$.
4. Is $x \mapsto |x|$ differentiable at 0?

Solution

1. $|x - 4| \leq 2 \iff 2 \leq x \leq 6$.
2. $2 - x \geq 0$, so the domain is $(-\infty, 2]$.
3. $(f \circ g)(x) = |x - 3|$.
4. No. The left slope is -1 and the right slope is 1 .

Limits, derivatives and integrals

Sequences: what is convergence?

Sequence convergence

A sequence (x_n) converges to L if its terms become arbitrarily close to L for large n .

Fast computation

For rational expressions in n , divide numerator and denominator by the dominant power of n .

Student task

Compute

$$\lim_{n \rightarrow \infty} \frac{5n + 2}{2n - 1}, \quad \lim_{n \rightarrow \infty} \frac{3n^2 - n}{n^2 + 4}.$$

Solution

$$\frac{5n+2}{2n-1} = \frac{5+2/n}{2-1/n} \rightarrow \frac{5}{2}.$$

$$\frac{3n^2-n}{n^2+4} = \frac{3-1/n}{1+4/n^2} \rightarrow 3.$$

Function limits: simplify before evaluating

Rule

If direct substitution gives $0/0$, factor or simplify first. Never cancel terms without remembering that the original expression is not defined at the cancelled point.

Student task

Compute $\lim_{x \rightarrow 1} \frac{x^2 - 1}{x - 1}$.

Solution

For $x \neq 1$,

$$\frac{x^2 - 1}{x - 1} = \frac{(x - 1)(x + 1)}{x - 1} = x + 1.$$

Therefore

$$\lim_{x \rightarrow 1} \frac{x^2 - 1}{x - 1} = 2.$$

Continuity

A function f is continuous at a if $\lim_{x \rightarrow a} f(x) = f(a)$.

Removable discontinuity

A removable discontinuity is a hole that can be filled by choosing the correct value.

Student task

Let $h(x) = \frac{x^2 - 4}{x - 2}$, $x \neq 2$. Compute $\lim_{x \rightarrow 2} h(x)$ and choose $h(2)$ to make the function continuous.

Removable discontinuity: solution

Solution

For $x \neq 2$,

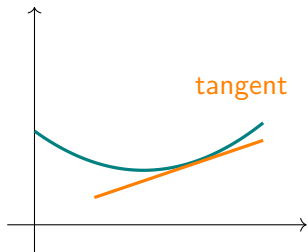
$$h(x) = \frac{(x-2)(x+2)}{x-2} = x+2.$$

Thus $\lim_{x \rightarrow 2} h(x) = 4$. To make h continuous, set $h(2) = 4$.

Derivative: slope and local variation

Derivative

The derivative $f'(a)$ measures the instantaneous slope of the graph at a .



Derivative rules used constantly

Function	Derivative
x^n	nx^{n-1}
e^x	e^x
$\log x$	$1/x$
$f + g$	$f' + g'$
fg	$f'g + fg'$
f/g	$(f'g - fg')/g^2$
$f(g(x))$	$f'(g(x))g'(x)$

Student task

Compute the derivative of

$$f(x) = x^3 - 4x + 1, \quad g(x) = e^{2x}, \quad h(x) = \log(x^2 + 1).$$

Solution

$$f'(x) = 3x^2 - 4, \quad g'(x) = 2e^{2x}, \quad h'(x) = \frac{2x}{x^2 + 1}.$$

Critical point

A point x_0 is critical if $f'(x_0) = 0$ or if the derivative does not exist there.

Student task

Find the critical points of $f(x) = x^3 - 3x$.

Solution

$$f'(x) = 3x^2 - 3 = 3(x^2 - 1).$$

The equation $f'(x) = 0$ gives $x^2 = 1$, hence $x = -1$ and $x = 1$.

Second derivative test

If $f''(x) > 0$ on an interval, then f is convex there. If $f''(x) < 0$, then f is concave.

Student task

For $f(x) = x^2 - 6x + 5$, find the minimum point and minimum value.

Solution

$$f'(x) = 2x - 6, f'(x) = 0 \iff x = 3.$$

Since $f''(x) = 2 > 0$, the function is convex and $x = 3$ is a minimum.

$$f(3) = 9 - 18 + 5 = -4.$$

First-order Taylor formula

Near a ,

$$f(x) \approx f(a) + f'(a)(x - a).$$

Meaning

Taylor approximation replaces a complicated function by a local line.

Student task

Use first-order Taylor approximation around 0 for

$$e^x, \quad \log(1+x), \quad (1+x)^2.$$

Solution

$$e^x \approx 1 + x, \quad \log(1 + x) \approx x, \quad (1 + x)^2 = 1 + 2x + x^2 \approx 1 + 2x.$$

The approximation keeps only constant and linear terms.

Integrals: accumulated quantity

Definite integral

The integral $\int_a^b f(x) dx$ represents accumulated signed area.

Why it matters later

In probability, integrals will become expected values for continuous random variables.

FTC

If $F(x) = \int_a^x f(t) dt$, then $F'(x) = f(x)$.

Student task

Compute $\int_0^2 (3x + 1) dx$ and find $F'(x)$ for $F(x) = \int_0^x (t^2 + 1) dt$.

Solution

$$\int_0^2 (3x + 1) dx = \left[\frac{3}{2}x^2 + x \right]_0^2 = 6 + 2 = 8.$$

By the fundamental theorem of calculus, $F'(x) = x^2 + 1$.

Five-minute check

1. $\lim_{n \rightarrow \infty} (2n + 1)/(n - 3).$
2. $\frac{d}{dx}(xe^x).$
3. Critical point of $x^2 + 2x.$
4. $\int_0^1 2x \, dx.$

Solution

1. The limit is 2.
2. $(xe^x)' = e^x + xe^x = e^x(1+x)$.
3. $2x + 2 = 0$, so $x = -1$.
4. $\int_0^1 2x \, dx = [x^2]_0^1 = 1$.

Linear algebra and multivariable calculus

Vector

A vector in $\mathbb{R}^n$ is an ordered list of n real numbers: $x = (x_1, \dots, x_n)$.

Operations

Vectors are added component by component and multiplied by scalars component by component.

Student task

Let $u = (1, 2, -1)$ and $v = (3, 0, 4)$. Compute $u + v$, $2u - v$, and $u \cdot v$.

Solution

$$u + v = (4, 2, 3), \quad 2u - v = (2, 4, -2) - (3, 0, 4) = (-1, 4, -6).$$

$$u \cdot v = 1 \cdot 3 + 2 \cdot 0 + (-1) \cdot 4 = -1.$$

Euclidean norm

$$\|x\| = \sqrt{x_1^2 + \cdots + x_n^2}.$$

The distance between x and y is $\|x - y\|$.

Student task

Compute the distance between $x = (1, 2)$ and $y = (4, 6)$.

Solution

$$x - y = (1 - 4, 2 - 6) = (-3, -4).$$

$$\|x - y\| = \sqrt{(-3)^2 + (-4)^2} = 5.$$

This is the two-dimensional version of $|x - y|$.

Matrices: always check dimensions

Matrix dimension

A matrix $A \in \mathbb{R}^{m \times n}$ has m rows and n columns.

Matrix product

The product AB is defined only if columns of $A =$ rows of B .

Student task

Let $A \in \mathbb{R}^{2 \times 3}$, $B \in \mathbb{R}^{3 \times 4}$, and $C \in \mathbb{R}^{4 \times 2}$. Which products are defined among AB , BA , BC , CA ? Give dimensions.

Solution

AB is defined and has size 2×4 .

BA is not defined because B has 4 columns and A has 2 rows.

BC is defined and has size 3×2 .

CA is defined and has size 4×3 .

Matrix-vector product

Each component of Ax is obtained by multiplying one row of A by the column vector x .

Student task

Compute

$$\begin{pmatrix} 1 & 2 \\ -1 & 3 \end{pmatrix} \begin{pmatrix} 2 \\ 1 \end{pmatrix}.$$

Solution

$$\begin{pmatrix} 1 & 2 \\ -1 & 3 \end{pmatrix} \begin{pmatrix} 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \cdot 2 + 2 \cdot 1 \\ -1 \cdot 2 + 3 \cdot 1 \end{pmatrix} = \begin{pmatrix} 4 \\ 1 \end{pmatrix}.$$

Linear dependence

Vectors are linearly dependent if one of them can be obtained from the others through a linear combination.

Student task

Are $v_1 = (1, 2)$ and $v_2 = (2, 4)$ linearly independent? Are $w_1 = (1, 0)$ and $w_2 = (0, 1)$ linearly independent?

Solution

$v_2 = 2v_1$, so v_1 and v_2 are linearly dependent.

For w_1, w_2 , the equation $aw_1 + bw_2 = (0, 0)$ gives $(a, b) = (0, 0)$, so they are linearly independent.

Determinant, rank and invertibility

Concept	Fast meaning
Determinant	For 2×2 matrices, $ad - bc$.
Invertibility	A square matrix is invertible iff determinant is nonzero.
Rank	Number of independent rows or columns.

Student task

For $A = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$, compute determinant, decide invertibility and rank.

Solution

$$\det A = 1 \cdot 4 - 2 \cdot 2 = 0.$$

So A is not invertible. The second row is twice the first, so there is only one independent row and $\text{rank}(A) = 1$.

Matrix form

A linear system can be written as $Ax = b$. If A is square and invertible, then $x = A^{-1}b$.

Student task

Solve mentally:

$$\begin{cases} x + y = 3, \\ x - y = 1. \end{cases}$$

Solution

Adding the two equations gives $2x = 4$, so $x = 2$. Then $x + y = 3$ gives $y = 1$. Thus $(x, y) = (2, 1)$.

Functions of two variables

Two-variable function

A function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ assigns one number to each point (x, y) .

Level curves

Level curves $f(x, y) = c$ are the two-dimensional analogue of reading the shape of a graph.

Gradient

For $f(x, y)$,

$$\nabla f(x, y) = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right).$$

Student task

Compute the gradient of $f(x, y) = x^2 + 2y^2 - 2x$.

Solution

$$\frac{\partial f}{\partial x} = 2x - 2, \quad \frac{\partial f}{\partial y} = 4y.$$

Therefore $\nabla f(x, y) = (2x - 2, 4y)$.

Stationary points in two variables

Student task

Find the stationary point of $f(x, y) = x^2 + 2y^2 - 2x$. Is it a minimum or a maximum?

Solution

Set the gradient equal to zero:

$$2x - 2 = 0, \quad 4y = 0.$$

Thus $(x, y) = (1, 0)$. Since the quadratic coefficients are positive, the function is convex and the point is a minimum.

Five-minute check

1. Dimension of AB if A is 5×2 and B is 2×7 .
2. Compute $(1, 1, 1) \cdot (2, 0, -2)$.
3. Are $(1, 1)$ and $(2, 3)$ dependent?
4. Gradient of $x^2 + y^2$.

Solution

1. AB is 5×7 .
2. $2 + 0 - 2 = 0$.
3. No, they are not multiples of each other.
4. $\nabla(x^2 + y^2) = (2x, 2y)$.

Statistics, probability and inference

Descriptive statistics: location

Index	Meaning
Mean	arithmetic center
Median	middle value after ordering
Mode	most frequent value
Quartiles	divide ordered data into four parts

Student task

For 2, 2, 4, 6, 6, compute mean, median and mode.

Solution

$$\bar{x} = \frac{2 + 2 + 4 + 6 + 6}{5} = 4.$$

The median is the middle value, 4. The modes are 2 and 6 because both appear twice.

Variance

For population data $x_1, \dots, x_n$,

$$\sigma^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2.$$

Student task

For 1, 3, 5, 7, compute range and population variance.

Solution

The range is $7 - 1 = 6$. The mean is 4. Therefore

$$\sigma^2 = \frac{(1 - 4)^2 + (3 - 4)^2 + (5 - 4)^2 + (7 - 4)^2}{4} = \frac{20}{4} = 5.$$

Covariance, correlation and regression

- ▶ Covariance measures joint movement.
- ▶ Correlation standardizes covariance and lies between -1 and 1 .
- ▶ A simple regression line has the form $\hat{y} = a + bx$.

Student task

If $r = -0.8$, describe the relationship. If $\hat{y} = 2 + 0.5x$, compute $\hat{y}$ at $x = 4$.

Solution

$r = -0.8$ indicates a strong negative linear relation.

At $x = 4$,

$$\hat{y} = 2 + 0.5 \cdot 4 = 4.$$

Probability space

A probability model is built on a sample space Ω , events $A \subseteq \Omega$, and a probability $P(A)$.

A^c	complement of A
$A \cap B$	both events occur
$A \cup B$	at least one occurs
$P(A \mid B)$	probability of A given B

Formula

If $P(B) > 0$,

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}.$$

Student task

Let $P(A) = 0.4$, $P(B) = 0.5$, and $P(A \cap B) = 0.2$. Compute $P(A \mid B)$. Are A and B independent?

Solution

$$P(A \mid B) = \frac{0.2}{0.5} = 0.4.$$

Since $P(A \mid B) = P(A)$, the information that B occurred does not change the probability of A . The events are independent.

Bayes formula

$$P(A \mid B) = \frac{P(B \mid A)P(A)}{P(B)}.$$

Student task

Let $P(A) = 0.5$, $P(B \mid A) = 0.8$, $P(B \mid A^c) = 0.2$. Compute $P(B)$ and $P(A \mid B)$.

Solution

$$P(B) = P(B \mid A)P(A) + P(B \mid A^c)P(A^c) = 0.8 \cdot 0.5 + 0.2 \cdot 0.5 = 0.5.$$

$$P(A \mid B) = \frac{0.8 \cdot 0.5}{0.5} = 0.8.$$

Random variable

A random variable is a numerical quantity whose value depends on the random outcome.

Discrete	takes isolated values, e.g. $0, 1, 2, \dots$
Continuous	takes values in intervals

Discrete expectation

If X takes values x_i with probabilities p_i , then

$$E[X] = \sum_i x_i p_i.$$

Variance

$$\text{Var}(X) = E[(X - E[X])^2].$$

Student task

Let $X \sim \text{Bernoulli}(0.3)$. Compute $E[X]$ and $\text{Var}(X)$.

Solution

For a Bernoulli variable,

$$E[X] = p, \quad \text{Var}(X) = p(1 - p).$$

With $p = 0.3$,

$$E[X] = 0.3, \quad \text{Var}(X) = 0.3 \cdot 0.7 = 0.21.$$

Standardization

If X has mean μ and standard deviation σ , then

$$Z = \frac{X - \mu}{\sigma}.$$

Student task

If $X \sim N(10, 4)$, standardize the value $X = 12$.

Solution

Here $\mu = 10$ and $\sigma = \sqrt{4} = 2$. Therefore

$$z = \frac{12 - 10}{2} = 1.$$

So $P(X \leq 12) = P(Z \leq 1)$.

Law of large numbers and CLT

LLN

Sample averages stabilize around the population mean when the sample size grows.

CLT

Properly normalized sample averages are approximately normal for large samples.

Common mistake

The CLT does not say that every random variable is normally distributed.

Confidence interval

An interval of plausible values for an unknown parameter, computed from data.

p-value

A measure of evidence against the null hypothesis. Small p-values indicate stronger evidence against the null.

Student task

Decide true or false:

1. If a 95% confidence interval is $[4.1, 5.3]$, the value 5 is compatible with it.
2. If a test has p-value 0.03 at significance level 5%, reject the null hypothesis.
3. The CLT says that every random variable is normal.

Solution

1. True: $5 \in [4.1, 5.3]$.
2. True: $0.03 < 0.05$.
3. False: the CLT concerns normalized sums or averages under suitable conditions.

Bridge to the first module

Prerequisite	Used immediately for
Absolute value	distance on $\mathbb{R}$
Intervals and neighborhoods	open balls and convergence
Sequences and limits	Cauchy sequences and completeness
Vectors and matrices	normed spaces and linear maps
Dot product	inner products and orthogonality
Variance and expectation	Hilbert space intuition later in probability

Before Module 1 starts

You should be able to:

1. solve simple absolute value inequalities;
2. compute domains, compositions and inverses;
3. compute basic limits, derivatives and integrals;
4. multiply matrices and vectors correctly;
5. compute mean, variance and simple conditional probabilities.

Next step: metric spaces

Distance, convergence, completeness and compactness